

GREEDYGAME: LEVERAGING ONLINE GAMING FOR BRAND STORYTELLING

Ashita Aggarwal, Suraj Commuri, and Ankit Rawal wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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During the summer of 2013, two engineering students in India, Arpit Jain and Arink Verma, noticed a rage of online gaming among their friends, peers, and young millennials. However, these ardent gamers were “players not payers;” while they loved playing for free, they were not keen to pay for those games. In India, a significant proportion of the population did not want to pay for the digital content that they consumed. The only option left for game developers, who had invested a considerable amount of money, was to monetize through advertisements (ads). However, unlike “lean back” forms of digital consumption where ads were acceptable, such as with watching a video or browsing the Internet, gaming was often a “lean forward” activity, and pop-up ads that were an eyesore in the beginning became interruptive and intrusive, spoiling the entire gaming experience. As such, a common practice for gamers was to reload or switch off the Internet while playing games to avoid pop-up ads.

Jain thought while ads were in some ways a necessary evil to subsidize content, there certainly must be a better way for a brand to seamlessly integrate into the content without affecting the user experience. Also, the ads needed to be relevant, and the gamer had to build a positive attitude for the brands advertised in the game. With this in mind, Jain and Verma took up the challenge of creating a sustainable gaming platform. Set up in 2015, GreedyGame Media Pvt. Ltd. (GreedyGame) enabled content creators to monetize their content through in-game advertising without disrupting the user experience. The online gaming as a product category was new; the users were unaware and did not know what to expect. GreedyGame had a massive task of creating a new category for gaming in India. In 2017, the founders set out to strengthen GreedyGame’s positioning and define a go-to-market strategy that could take GreedyGame’s mobile advertising from mindshare to market share in India and beyond.

THE GAMING INDUSTRY: AN OVERVIEW

Online games were played using consoles, personal computers (PCs), laptops, or mobile handsets. They were accessed through online channels, such as a browser; were preloaded or application (app)-based; and required Internet access in primary gameplay experience and for monetization. Online games could be played across single-single, multi-player, and massively multi-player formats. Due to its spread across wide genres and its application across business sectors, gaming was attracting interest from private equity and strategic investors. According to a study undertaken by KPMG and Google Inc. (Google), the online gaming

sector was set to add 190 million gamers and to grow from US\$290 million¹ in 2018 to a \$1 billion opportunity by 2021.² According to a report by Ernst & Young LLP and FICCI, “Mobile was the platform of choice for gamers, contributing nearly 87% of gamers in 2017.”³

With its digital gaming industry growing rapidly, India was poised to become one of the top five mobile gaming countries in the world. In 2010, only 25 countries had a digital gaming industry; by 2018, that number increased tenfold to 250. Many middle-class Indians were engaging in digital gaming platforms through console and PCs.⁴ By 2005, social media had introduced Indians to online gaming. The adoption of social games was facilitated by international developers. The gaming experience required expensive gaming consoles and PCs, which had traditionally kept gaming out of reach for the majority of Indians. Mobile phones democratized gaming and moved it from consoles and PCs to handsets.

India had shown exponential growth, with a rise in both Internet penetration and the number of smart phone users. Internet penetration was 31 per cent in 2017 and was expected to reach 53 per cent by 2021.⁵ Smart phone users were projected to reach 470 million by 2021.⁶ This surge was expected to be complemented by a fivefold increase in data consumption⁷ coupled up with an overall increase in disposable income. Gaming, which had previously been accessible to only a privileged few, was now available to a much wider demographic. In 2016, India was among the top five countries for mobile game downloads across iOS and Android stores. Gaming became the most popular mobile activity for an average Indian consumer after messaging and mobile video consumption. As low-cost Android devices across lower-tier cities increased, the bulk of growth in both downloads and revenue was driven by Google Play. Google Play generated 13 times more downloads compared to the iOS App Store.⁸

The development of comprehensive local expertise, the entrepreneurial nature of Indian consumers, and the influx of fresh funds led Indian companies to grow from being merely service providers to becoming end-to-end game developers who could customize the content to local market preferences. The number of local service providers increased from five in 2005 to more than 25 by 2010.⁹ Games based on the local language and local themes were able to connect with users and hence positively affected consumption patterns. Digital payment users were set to increase to 300 million by 2020.¹⁰ These changes were further exacerbated by new technologies, such as virtual reality, which were available at affordable rates (see Exhibit 1).

The gaming industry was divided into three categories: casual games, social games, and real-money games. Monetization was realized through revenue streams such as in-app advertising (based on cost per click and cost per mile), in-game brand placements, pay per download, subscription services, in-app purchases (using virtual currency), freemium upgrades, and incentive-based ads. Indian consumers were still apprehensive of paying for any digital content. Therefore, most developers were dependent on advertisers and publishers for monetization. As the market grew, it was expected to move to a freemium model to recognize a better balance of revenue realization from the ecosystem and online gamers.¹¹

It was expected that until 2017, online gaming would be driven by ecosystem-led monetization due to low volumes and usage. This included in-app ads by brands (banner ads or pop-up ads displayed during games), in-game brand or product placements (popular brands or products placed in games and integrated with the gaming experience), or incentive-based ads (virtual objects or access to the game after watching ads). As gamer volume and usage was expected to increase over the next five years, the industry would move to a freemium model and gamer-led monetization, where the user was ready to pay for the gaming experience and enhancements. These included pay-per-download offers (upfront charges for access to the game or the download), upgrades (a free download, but with charges applied for full access to the game’s features, characters, and levels), in-app purchases (to enhance the experience of gaming by paying for advanced products and features), or the subscription model (an annual or monthly package for extended access after the trial period was over).

There was a clear relationship between downloads, usage, and revenues. The United States was a mature market and was in the middle of the shift from the download growth phase to app usage and revenue. As an emerging market, India was still experiencing growth in app downloads. App store revenue from games grew by 80 per cent between 2014 and 2016. Gamers on iOS contributed 70 per cent of combined iOS App Store and Google Play store revenue in 2014. Data suggested that contribution was higher from premium segments.¹² However, with Android phones challenging Apple Inc.'s supremacy in the premium smart phone segment, Google Play's app spend was expected to increase in the coming years. In India, racing and action categories dominated downloads. However, strategy and social card games contributed the most to revenues. Consumers were spending the most time on strategy and casual games, with *Candy Crush Saga* securing the top spot.¹³

An awareness of Indian culture and beliefs was essential to designing games that could connect to a larger number of users in India. One of the reasons why social games worked well was because they were similar to an old card game played by Indians—*Teen Patti*. Similarly, mobile cricket games were adopted more than games featuring other sports due to the emotional connection that Indians had with cricket.

The Indian Online Gamer: Decoding Consumer Behaviour

The average Indian online gamer was a 24-year-old male who had been introduced to gaming by his family or peers. Of urban mobile gamers, 60 per cent were below the age of 24, although experienced working professionals and stay-at-home consumers were almost as likely to play online games. However, 80 per cent of gamers were male, and this skewed distribution left an opportunity for penetration in the Indian market.¹⁴ Approximately 48 per cent of gamers surveyed by Nielsen had tried online gaming after watching a peer group or after receiving word-of-mouth references.¹⁵ More than 64 per cent of first-time gamers had been influenced by friends, family, or peer groups to download a game. Playing for recreation or making use of idle time was the second-most common reason for initiating online gaming. More than 75 per cent of gamers were continually engaged with online gaming because they perceived it as a medium of stress relief or as a means for social interaction. More than 29 per cent preferred puzzle, action, and adventure games, and 43 per cent of downloads took place through app distribution platforms.¹⁶ However, many potential gamers were hesitant to indulge in online gaming due to high data consumption, a negative perception among peers and family, and a deterioration in device performance. Gamers perceived paid games as being better than freemium games.¹⁷

Heavy gamers spent about 33 per cent of their online time on games, followed by communication channels, and consumed up to 800 megabytes (MB) of data on games alone. This segment was dominated by males (87 per cent) under the age of 25. Of this segment, 88 per cent accessed games on their mobile phones and engaged in various genres, spending an average of more than 30 minutes per day on online gaming. Compared to this, casual gamers spent only about 7 per cent of their online time on games, thus allowing time for social communication (25 per cent of total time spent online) and online browsing. This segment consumed about 188 MB of data for online gaming and spent less than an average of 30 minutes per day on gaming. 20 per cent of casual gamers were female.¹⁸

According to research conducted by KPMG and Google in 2017, heavy and casual gamers showed similar preferences for gaming genres, including puzzle, action, and adventure games. According to the research, female players preferred puzzles, while male players preferred sports, action, and social card games. Adventure and strategy games found similar resonance across genders.¹⁹ The urban smart phone gamer indulged in multiple activities, of which chats and VoIP (Voice over Internet Protocol) constituted 22 per cent of the total time spent online, followed by online browsing (18 per cent) and online gaming (15 per

cent). Indian gamers, on average, spent 11 per cent of their entire entertainment wallet on online gaming. A heavy player was spending 46 per cent more time than others on online entertainment. A gamer's payment behaviour was related to the perceived value of online gaming. For non-paying online players, barriers to adoption included the availability of free substitutes (30 per cent of gamers), perceived high pricing (26 per cent), perceived waste of money (25 per cent), reduced competition in the game (14 per cent), and a lack of comfort with online payment (12 per cent). Even for paid gamers, payment preferences were skewed toward credit or debit cards. Students and stay-at-home consumers preferred non-banking payment modes such as mobile wallets. Payment mechanism was a reflection of income group and financial independence (see Exhibits 2 and 3).²⁰

Gaming in Business

The top 100 freemium games generated 22 times more revenue in comparison to the top 100 paid games. In 2016–17, there were 392.5 million downloads for freemium games, generating \$19.65 million. Paid games contributed \$0.87 million through 231,000 downloads. Advertisers and publishers dominated the monetization avenues. As most gamers felt that prices for online games were not affordable, the in-app purchases offered flexibility to keep costs low. Among the top 50 games by revenue, strategy, puzzle, and chance-based games dominated the total revenue and average revenue per game.²¹ In 2016–17, the top three games by downloads in freemium and paid models were *Subway Surfers*, *Candy Crush Saga*, and *Temple Run 3*. The top three games by revenue were *Clash of Clans*, *Clash of Kings*, and *Candy Crush Saga*.²²

Online gaming was not just an entertainment offering; it was slated to become a serious business, as it could be used by businesses to connect and engage with consumers. Gamification (i.e., the application of gaming mechanisms in non-game situations) was increasingly being used as a business tool to enhance operations and increase customer and employee engagement levels. Gamification was gradually merging with enterprise technology solutions and tools such as customer relationship management. Though less than 10 per cent of India's organizations was using gamification in 2017, it was slowly finding its way into the marketing budgets of brands. E-commerce introduced games such as spin wheels, bid prices, and scratch cards to engage with customers; technology companies were engaging with new recruits using interactive and incentive-based learning; and many educational portals were using games, simulations, and videos as training tools.²³ Using gamification as a brand-building tool was a great opportunity for gaming companies in India. It provided firms with an opportunity to expand their portfolios and include additional revenue models. The success of this model rested in building alliances and partnerships, as the ecosystem was huge, and capabilities were diversified.

GREEDYGAME: PLAYING IT DIFFERENTLY

In 2013, two bright students at the Indian Institute of Technology Ropar—Jain (26 years of age) and Verma (24 years of age)—were working together on a project called Enactus to build a sustainable business model. The two students did not find the school project particularly interesting, but they enjoyed working together and were mutually interested in the popularity of mobile gaming, including among their own friends. They observed that users would switch off their Internet while playing games to avoid pop-ups and banners. This incidence sparked the idea to start a platform driven by content but featuring the seamless integration of ads, to avoid spoiling the user experience. And that was the beginning of GreedyGame.

Challenges

Before embarking on the project, Jain and Verma were apprehensive of the project's ability to generate revenue. Research showed that banners and interstitial ads were intrusive and annoying for users who were highly involved in playing a game. Click-through rates were less than 0.5 per cent; therefore, these forms of advertising were not a lucrative option for advertisers. The duo took on the mission to make ads look beautiful within the online games and to narrate the brand story to enhance the user's gaming experience. They created medium-agnostic advertising that catered to content by creating a self-serve platform that blended ads within free content natively. This kind of a platform enabled non-intrusive monetization through which publishers could sustain themselves. It also helped users to get what they wanted: uninterrupted, free content. GreedyGame discovered that 70 per cent of individuals wanted to learn about products or content through branded content, rather than through traditional advertising. Branded content enabled advertisers to be a part of gaming consumption, enhancing their visibility and reach.

The current generation of gamers could be aptly described as a skip generation. These customers sought high-quality content; however, content creators could not monetize this content because these users wanted content for free and did not have the time or inclination to consume advertising as separate content. As the online model moved to a consumer-to-consumer model, there was more content than time to consume. One study conducted by GreedyGame found that, in 60 seconds, 72 hours of video content was uploaded, 2.4 million posts were shared on Facebook, 48,000 apps were downloaded, and 4.8 million gaming sessions ran concurrently. Armed with this information, GreedyGame knew that traditional content and gaming models would not work for long.

The Approach

Jain and Verma, developed a platform that was driven by content, as opposed to media (such as those developed by Yahoo or Google). Therefore, the platform used engaging units of the game such as billboards, storefronts, characters, and objects, integrating the ads intelligently and intuitively. GreedyGame followed an approach wherein content creators or publishers integrated their branding into the content so that ads were seamlessly blended into the content, with no distractions for users. The ads were contextual and targeted users. Publishers could publish content on any platform and were paid for ads based on tracked key performance indicators specific to the platform.

To prove that this new approach was effective, GreedyGame started building games with integrated advertising. Users did not feel that branding was an intrusion; they found that they could engage with the ads. This increased the connection between both the platform and the brands. The brand advertisers could specify the target audience to GreedyGame to filter content that suited the target segment's needs based on demographics, location, purchase behaviour, and so on. They would then select content from among formats, including games, images, videos, and virtual reality, to check how the brand blended into the content. Advertisers also benefited because they then paid only for ad views only, and not for the content.

To address the concern of content overload and growing user dissonance, GreedyGame had to innovate and develop content that was not only engaging and free, but also blended with the ads natively. The medium-agnostic advertising could cater to the content and enabled publishers to monetize through non-intrusive advertising. Advertisers could become a part of consumption and enhance their visibility and reach. Finally, uninterrupted free content could be made available to users. The branded clickable unit was introduced to support more messaging and videos, which increased the click-through rate to 1.5 per cent. GreedyGame incorporated native ad placement in games in a plug-and-play manner using billboards and power-ups. A two-click process was initiated to avoid accidental clicks (see Exhibits 4 and 5).

The Strategy at Launch

Jain and Verma knew that the GreedyGame business could be monetized only through ads. Companies approached GreedyGame when they had gaming properties that could engage people across genders and age groups. The intended user encompassed anyone between the ages of three and 60 who was interested in online gaming. Any organization or brand seeking to engage with its consumers was considered a potential GreedyGame customer. GreedyGame also developed a deep understanding of its intended target customers by studying device characteristics, real-time activity, behavioural traits (e.g., user interests, purchase history, etc.), and brand affinity.

To grow the company faster, Jain and Verma decided that instead of building their own games, they would license their technology for use by other game developers and publishers, and monetize their work through relevant native ads. The team started working on all formats to advertise brands within games, including in-game non-intrusive placements, rich media content, in-game UI (user interface) videos, integrated brand surveys, and customized brand games. This helped them scale faster. By 2017, GreedyGame had reached 75 million gamers in India per month and was able to show ads in more than 200 games across several genres, including cricket, racing, cards, and puzzles. Users were targeted by relevant brands that were seamlessly integrated into the game's storyline.

One competitive advantage GreedyGame had was the built-in context-agnostic technology. Gaming suffered from a low shelf life, as gamers became bored quickly and were always looking for new games. The retention rate for games was less than 5 per cent per month (95 per cent of mobile games were churned within a month). As such, game publishers and developers aspired to make a blockbuster (such as *Angry Birds*, *Candy Crush Saga*, or *Subway Surfers*) by regularly releasing several titles a year in the hope that something would stick with users. Therefore, the technology that powered ads in these numerous games had to be gaming-content agnostic. The bigger the game in terms of monthly active users, the higher reach and, therefore, the higher the monetization potential.

The platform charged by the number of sessions per game, including the cost per session. A session started after the user crossed 60 seconds of gameplay, with the platform usually charging between \$8 and \$10 for every 1,000 sessions. However, the pricing could differ based on internal analytics of user interaction and engagement as well as the complexity of the game. Any ad technology platform was a two-sided network with demand and supply.

Demand represented the party who was paying for the ad. Typically, an advertiser, brand, or media agency constituted the demand-side platform. The content where the ad showed up, such as through the game and its publishers, represented the supply-side platform. Bigger companies like Google, Facebook Inc., and InMobi played on both sides of demand and supply. However, doing so was expensive, as it required huge capital, technical know-how, and management bandwidth to cover end-to-end value chains that enabled companies to capture a larger share of the pie. Smaller companies were constrained by resources and focused only on one side to extract a higher value. GreedyGame developed an SSP (Self Service Portal), software-as-a-service, and affiliate revenue model (see Exhibit 6).

GreedyGame was a young start-up, and funding was a challenge. As the business grew, there was a growing need for resources. In September 2015, the firm raised a seed fund of \$200,000 from Times Internet and other well-known angel investors. They raised funds worth \$20,000 over two rounds and were actively using 19 technologies for their website. However, they still required \$6 million over the next year to tap into new technology, develop new offerings, and expand the market.

Competition for GreedyGame

GreedyGame was competing at multiple levels, with companies involved in content- and medium-driven advertising. Gaming companies such as POKKT, Gameloft SE (Gameloft), and Unity Ads showed ads on gaming platforms, and GreedyGame helped these companies monetize games and content through ads. It also had advertising technology companies such as Google, Facebook, and InMobi competing for advertisers' budgets on digital platforms. However, traditional media, such as television and radio, were also competing for advertisers' money. With annual revenue of \$4.5 million, GreedyGame was competing with both traditional media and well-entrenched digital players, all vying for the media budget of brands (see Exhibit 7).

Direct Competition for GreedyGame

POKKT

POKKT was a leading mobile video advertising platform in India, South East Asia, and the Middle East and North Africa region with a strong focus on rewarded video ads within mobile games. POKKT was integrated with over 300 top games and app developers. By integrating POKKT platforms, developers could monetize their non-paying users to maximize online CPMs (cost per thousand users) through highly targeted brand video ads within their apps. Their key offerings for brand advertisers included highly effective targeting, opt-in native video ads, completed views and exclusive games inventory in India, and search engine advertising.²⁴

Nazara Technologies

Nazara Technologies was a leading diversified gaming and sports media platform based in India. With a presence in India and across emerging and developed global markets such as Africa and North America, the platform offered interactive gaming, e-sports, and gamified early learning ecosystems. The company boasted some of the country's most recognized gaming titles. Its mobile games included *World Cricket Championship*, *Carrom Clash*, and *Chhota Bheem*, which was based on a comical character in an Indian children's book. Its gamified early learning titles included *Kiddopia*, an online game for children. Its e-sports titles included *Nodwin* and *Sportskeeda*. Skill-based fantasy and trivia games included *Halaplay* and *Qunami*. The company's strengths were concentrated mainly in the children's entertainment segment and in online gaming.²⁵

Gameloft Advertising Solutions

Gameloft Advertising Solutions was founded in 1999 and produced games for consumers to download to their mobile phones. As a leading mobile game publisher, the company featured titles such as *Asphalt 8*, *Dragon Mania Legends*, *Little Big City*, *Sonic Runners*, and the *Modern Combat* series. It relied on its established and known expertise and resources to design gaming experiences and create meaningful connections between gamers and brands.²⁶

Unity Ads

Unity Ads was a two- and three-dimensional game development platform that served as a comprehensive monetization platform for app developers on the Unity, iOS, and Android operating systems. It helped companies integrate relevant ads and in-app purchases into the gaming experience to increase earnings.²⁷

Adjacent Competitors

InMobi

InMobi was a leading technology company that specialized in mobile advertising and marketing. Its mobile-first platform allowed brands, developers, and publishers to engage consumers through contextual mobile advertising. The company's platform aimed to bring together publishers and advertisers to provide relevant, personalized, and contextual ads to app and website users. InMobi worked closely with over 30,000 app developers across gaming, social, news, utility, and entertainment verticals, among others, to help monetize their apps using mobile advertising.²⁸

Affle

Affle was a global technology company that delivered consumer engagement, acquisitions, and transactions on its proprietary consumer intelligence platform. The platform aimed to enhance returns on marketing investment through contextual mobile ads and by reducing digital ad fraud. Affle's consumer platform was used by both online and traditional companies for measurable mobile advertising. The company helped traditional companies go online through its platform-based app development and a customer data service.²⁹

Google AdMob

Google AdMob helped monetize mobile apps through in-app ads.³⁰ According to GreedyGame, Google AdMob competed against GreedyGame for supply partnerships. However, GreedyGame focused on native advertising, a format that was not available on Google AdMob.

Facebook Audience Network

Facebook Audience Network offered in-app monetization to third-party apps, in addition to Facebook-owned apps (such as Instagram).³¹ According to GreedyGame, Facebook Audience Network (like Google AdMob) had a global pool of mobile game developers. Games made in one part of the world were readily made available to gamers across different geographies. Although the company did not have a native-only ad format, it was the first choice of game developers and publishers due to its experience with user acquisition and monetization.

THE RESULTS

By the end of 2017, GreedyGame was doing business with more than 31 publishers and 25 advertisers. The company was reaching 25 million unique gamers out of 1.5 billion gamers globally. GreedyGame had almost 1.5 million unique daily users, with more than three million gaming sessions. The company integrated over 60 games, with an average engagement rate of 8 per cent and an average click rate of 1.5 per cent. GreedyGame's inventory grew 70 per cent per month, with monthly revenue growth of 40 per cent. The company was a direct advertiser for Marico Limited, Flipkart, and Quickr. It also worked with

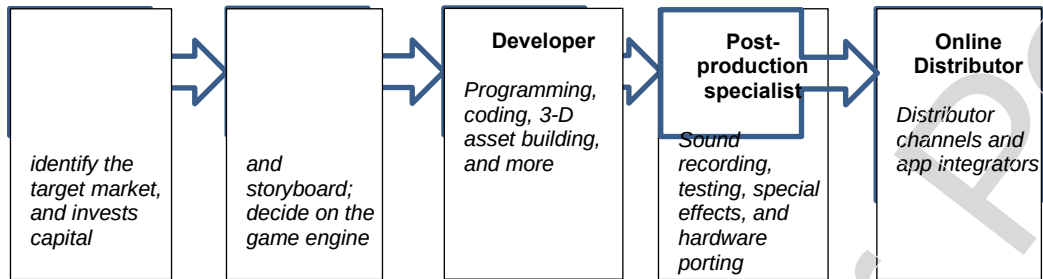
agencies such as GroupM to offer its services to companies such as Airtel, Titan by Tata, Pepsico Inc., and Tata Group. GreedyGame also partnered with Dentsu Aegis Network (whose clients included Mondelez International Inc., Acer Inc., and Adidas AG) and Omnicom Media Group (for brands including Boost and McDonald's). The company worked on building gaming content for Dettol, ICICI Bank, Amazon.com Inc., Vodafone Group PLC, Vivo Communication Technology Co. Ltd., and OnePlus Technology Co. Ltd, among others, and worked on launch campaigns with Godrej Group, Pizza Hut, and Myntra.

By the end of 2017, GreedyGame had achieved various critical milestones. The company could create technology to blend ads within mobile games in real time. It had achieved 60 per cent automation for campaign creation via in-house image processing. It had also partnered with top gaming publishers in India including Octro Inc., Nazara Technologies, Reliance Games, and Games2win. It had introduced a new model of costing that was more suited to the content than the medium. GreedyGame had managed to generate steady revenues and quality traffic from many countries and regions including India, the United States, the Middle East, and Pakistan. These achievements were critical to earn repeat campaign business.

GreedyGame was entering an expansion phase and hence wanted to further crystalize its value proposition and expand its reach and business. The capital raised by the company was getting exhausted, as the expenses were still higher than the revenue. There was intense competition from larger, well-funded, and entrenched players. However, Jain's biggest challenge was changing the evaluation lens of the stakeholders (customers, investors, and employees) to create a new category and an innovative technology solution. Brand managers were still apprehensive about the reach and effectiveness of native ads within games and therefore allocated limited marketing budgets toward game advertising. With the online gaming category being new, there was not enough evidence, data, or case studies by which to measure marketing effectiveness.

With these concerns as the backdrop, Jain called for a meeting with his board to put together a plan for 2018–2020 that could help in long-term sustenance while patiently building an online gaming category and waiting for his “moment of victory.” Before heading into the next decade, he wanted to strengthen GreedyGame's positioning and define a go-to-market strategy that could take GreedyGame's mobile advertising from mindshare to market share in India and beyond. Game on!

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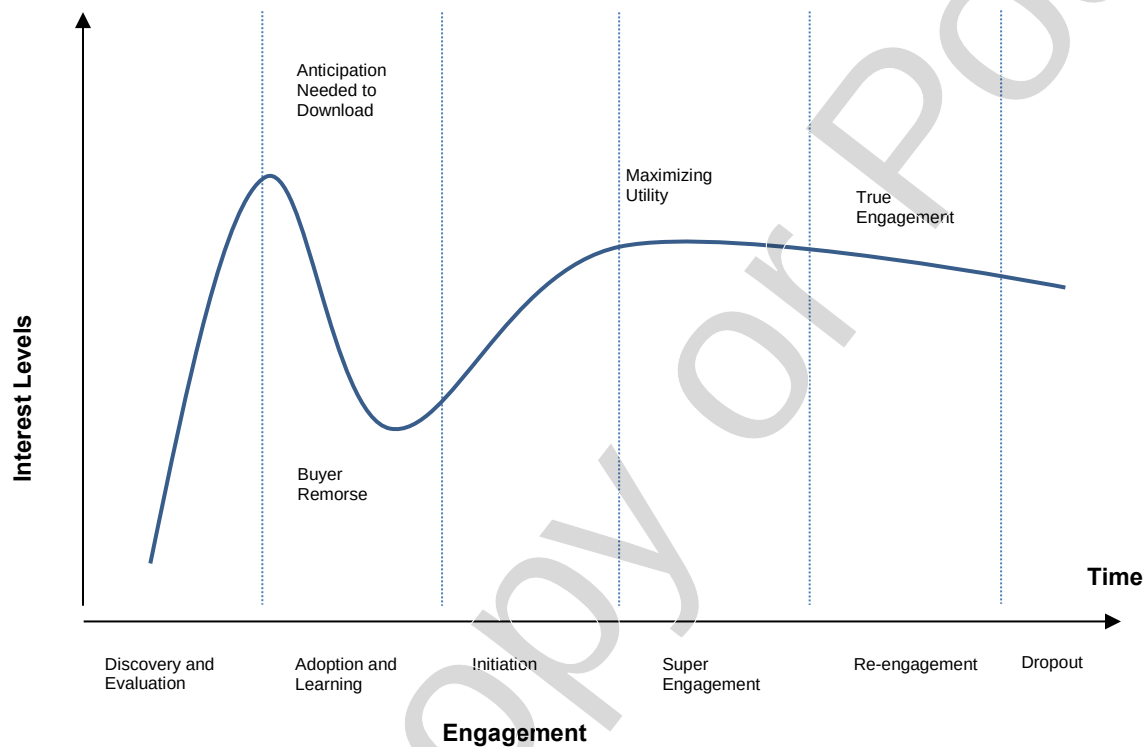
EXHIBIT 1: THE GAMING CONCEPT AND ECOSYSTEM

Support Network	
Device OEMs	Mobile, PC, laptop manufacturers
Advertiser and Advertising Platform	Advertisers on gaming platforms
Internet Service Providers	Enable distribution and consumption of online games
Type of Players	
Integrated Players	Publishers involved across the development cycle with resources and capabilities to perform all activities in the value chain
Co-publishers	Along with the publisher, co-publishers investing in the development of a game and providing one or more services along the value chain
Ancillary Service Providers	Providing one or more services during development of the game, such as testing, hardware, and animation

Note: 3-D = three-dimensional; OEM = original equipment manufacturer; PC = personal computer..

Source: KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, p. 15, May 2017, accessed October 26, 2019, <https://assets.kpmg/content/dam/kpmg/in/pdf/2017/05/online-gaming.pdf>.

EXHIBIT 2: A GAMER'S LIFECYCLE



Source: "Analytics & the Player Lifecycle," *Unity Technologies* (blog), accessed February 2, 2020, <https://blogs.unity3d.com/2016/05/03/analytics-the-player-lifecycle/>.

EXHIBIT 3: TRIGGERS AND MOTIVATORS FOR ADOPTION

Discovery—Motivators	Selection and Download—Motivators	Play and Engage—Triggers	Dropout—Triggers
Word of mouth, 53%	Popularity of game, 68%	Destress, recreation, 40%	High data consumption, 42%
Friends, family, and peers, 56%	Friends, family, and peers, 63%	Social interaction, 38%	Negative perception, 37%
Advertisements (offline and online), 40%	Game demo on app, online videos, and distribution platform, 55%	Sense of achievement or competitive spirit, 34%	Reduce device performance, 36%

Source: KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, May 2017, accessed October 26, 2019, <https://assets.kpmg/content/dam/kpmg/in/pdf/2017/05/online-gaming.pdf>.

EXHIBIT 4: HOW THE GAMING ECOSYSTEM WORKS FOR CREATORS AND PUBLISHERS

STEP 1	STEP 2	STEP 3
Create content that users like to consume: - No worry about ad placements - No need to make content focused on brand's needs	Integrate the platform to enable branding opportunity within the created content: - Ads seamlessly blended into the content - No distraction for users - Contextual ads shown to targeted users	Publish content on any platform: - Get paid for ads placed based on key performance indicators specific to the platform - Real-time dashboard to track monetization

Note: ad = advertisement

Source: Created by the authors based on company data.

EXHIBIT 5: HOW THE GAMING ECOSYSTEM WORKS FOR BRAND ADVERTISERS

STEP 1	STEP 2	STEP 3
Specify target group to filter content that suits advertiser's needs of - location; - demographics; - purchasing power; - behaviour; and - content genre.	Select content from a large array of recommendations of various formats (games, image, video, virtual reality). Check how brand blends into the content.	Pay for views and not the content. Define key performance indicators, track in real time. Pay only for users who consume the content.

Source: Created by the authors based on company data.

EXHIBIT 6: GREEDYGAME REVENUE MODELS

As an Ad Network	As a Supply-Side Platform	As a Software-as-a-Service Platform
Focus on both demand and supply, trying to build two-sided networks. - Aggregate both demand and supply. - Work with brands as advertisers, on one side, and gaming publishers, on the other. Realized quickly that deep pockets and big technical chops were needed. With big entrenched players in the game, the competition was intense.	Focus only on the supply side (i.e., gaming) go deeper than the end-to-end approach. - Aggregate different genres of games. - Monetize these games through native ads. - Give minimum guarantees (fixed revenue) or percentage of revenue to game publishers to monetize their games. While better than a two-sided network, this system was still susceptible to supply-side shocks. Also, since the native advertising was yet not evolved and the monetization models were not clear, there was resistance among game developers to take the software development kit. A lot of time and money went to onboarding a big game publisher to the platform.	Focus on charging a fee for using its native ads technology. This is similar to a software company charging usage or licensing fees. - Limits any demand- or supply-side shock. - Assures that some revenue comes from game publishers for using the technology. - Uses the pay-as-you-go subscription model. - The number of clients is reduced drastically. This model helped GreedyGame stay afloat but stunted growth, as very few game publishers were open to sharing a monthly fee.

Note: ad = advertisement.

Source: Created by the authors based on company data.

EXHIBIT 7: COMPETITION ANALYSIS FOR GREEDYGAME

	Performance and Direct Response Advertising	Brand Advertising
Content-Driven Advertising	• GreedyGame • Gameloft SE • Unity Ads • POKKT • Zero Games Studios • The Viral Fever • Pocket Aces	Owned and Operated Digital Networks: • YouTube • Facebook • Instagram • Gaana.com • New Delhi Television Ltd. • Dailyhunt • Disney+Hotstar • ZEE5
Medium-Driven Advertising	• Google AdMob • Facebook Audience Network • InMobi • Affle • Chartboost • Vungle • AdColony • AppLovin	Traditional Players: • Television • Print • Radio • Out of Home Media

Source: Created by the authors based on company data.

ENDNOTES

- ¹ All currency amounts are in US dollars unless otherwise specified.
- ² KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, May 2017, accessed October 26, 2019, <https://assets.kpmg/content/dam/kpmg/in/pdf/2017/05/online-gaming.pdf>.
- ³ Ernst & Young LLP and FICCI, *Re-imagining India's M&E Sector*, March 2018, accessed February 5, 2020, <http://ficci.in/spdocument/22949/FICCI-study1-frames-2018.pdf>.
- ⁴ Suparna Dutt D'Cunha, "How Digital Gaming in India Is Growing Up into a Billion-Dollar Market," *Forbes India*, March 9, 2018, accessed October 4, 2020, www.forbes.com/sites/suparnadutt/2018/03/09/how-online-gaming-in-india-is-growing-fast-into-a-billion-dollar-market/?sh=6be9710355b6.
- ⁵ KPMG in India and Google, *Indian Languages—Defining India's Internet*, April 2017, accessed February 6, 2020, <https://assets.kpmg/content/dam/kpmg/in/pdf/2017/04/Indian-languages-Defining-Indias-Internet.pdf>.
- ⁶ Arun Gupta, "Future of Contesting and Gamified Online Quiz Streaming Apps," *BusinessWorld*, July 7, 2018, accessed December 20, 2019, www.businessworld.in/article/Future-Of-Contesting-And-Gamified-Online-Quiz-Streaming-Apps/07-07-2018-153759/.
- ⁷ Ericsson, *Ericsson Mobility Report: On the Pulse of the Networked Society*, June 2016, accessed December 17, 2019, www.ericsson.com/en/mobility-report/reports.
- ⁸ Donny Kristianto, *Mobile Gaming on the Rise in India: Market Intensifies as Gamer Population Surges*, accessed March 2016, www.animationxpress.com/wp-content/uploads/2016/11/Mobile_Gaming_India_EN_17112016.pdf.
- ⁹ KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, op. cit.
- ¹⁰ Boston Consulting Group and Google, *Digital Payments 2020: The Making of a \$500 Billion Ecosystem in India*, July 2016, accessed October 26, 2019, https://image-src.bcg.com/BCG_COM/BCG-Google%20Digital%20Payments%202020-July%202016_tcm21-39245.pdf.
- ¹¹ KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, op. cit.
- ¹² Kristianto, op. cit.
- ¹³ Ibid., 10.
- ¹⁴ "Player Profile: Indian Online Gamer Statistics," News Karnataka, November 14, 2019, accessed October 14, 2019, www.newskarnataka.com/sports/player-profile-indian-online-gamer-statistics.
- ¹⁵ KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, op. cit.
- ¹⁶ Nimisha Jain and Kanika Sanghi, *The Rising Connected Consumer in Rural India* (The Boston Consulting Group, July 2016), accessed October 24, 2019, https://image-src.bcg.com/Images/BCG-The-Rising-Connected-Consumer-in-Rural-India-July-2016_tcm9-61868.pdf.
- ¹⁷ KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, op. cit.
- ¹⁸ Abhijit Matkar and Ayan Roy Choudhary, *What Handsets Say about Consumers* (The Nielsen Company, September 2018), accessed October 26, 2019, www.nielsen.com/wp-content/uploads/sites/3/2019/04/nielsen-featured-insights-what-handsets-say-about-consumers.pdf.
- ¹⁹ KPMG in India and Google, *Online Gaming in India: Reaching a New Pinnacle*, op. cit.
- ²⁰ Ibid.
- ²¹ Ibid., 15.
- ²² Kristianto, op. cit.
- ²³ Ernst & Young LLP and FICCI, op. cit.
- ²⁴ "About Us," Pokkt Mobile Ads, accessed February 16, 2021, www.pokkt.com/about.
- ²⁵ "We're Nazara!" Nazara Technologies, accessed February 16, 2021, https://corp.nazara.com/?page_id=2175.
- ²⁶ "World Presence," Gameloft Advertising Solutions, accessed February 16, 2021, www.gameloft.com/corporate/en/company/world-presence.
- ²⁷ "Grow your business with Unity Ads," Unity Ads, accessed February 16, 2021, <https://unity.com/solutions/unity-ads>.
- ²⁸ "Our Story," InMobi, accessed February 16, 2021, www.inmobi.com/company.
- ²⁹ "About Us," Affle, accessed February 16, 2021, <https://affle.com/about-us>.
- ³⁰ "Earn More Revenue with Your Apps," Google AdMob, accessed February 16, 2021, <https://admob.google.com/home>.
- ³¹ "Audience Network," Facebook, accessed February 16, 2021, www.facebook.com/audienccnetwork.